



Engineer to Excel

BANK CUSTOMER DATA ANALYTICS USING CORRELATION AND COVARIANCE MEASURES



Data-driven insights for customer understanding, risk analysis & banking decisions

SCENARIO

Bank customer data including age, income, account balance, transactions and loan amount is analyzed using correlation and covariance to identify relationships and customer patterns.

A I B T L
Age Income Balance Transactions Loan

1 DATA SUMMARY OF BANK CUSTOMER PATTERNS

•Key customer measures show the typical profile and variation in the dataset.

Average Age: 36 years
Average Income: ₹68K
Average Balance: ₹1.25L
Average Loan: ₹2.8L
Transactions: 18/month

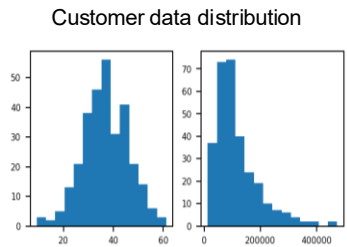


KEY INSIGHTS
• Income ↔ Balance: positive
• Transactions vary by segment

1,800 CUSTOMERS

2 ANALYSIS OF BANK CUSTOMER DATA DISTRIBUTION

•Distribution analysis identifies typical values, variability and unusual financial profiles.

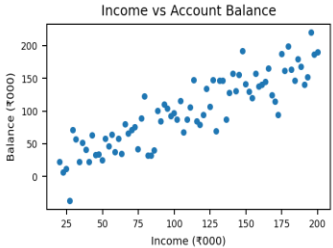


Right-skewed: income & balance

3 IDENTIFICATION OF UNUSUAL CUSTOMER PROFILES

•Unusual profiles occur when financial values differ strongly from the typical customer range.

HIGH-VALUE CUSTOMER
Balance: ₹8.5L



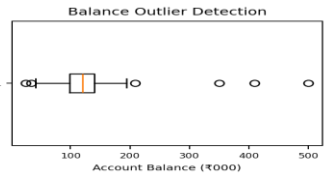
4 OUTLIER DETECTION & TREATMENT

DETECTION

- IQR method
- Z-score method
- Percentile method

OUTLIERS

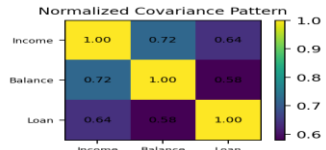
Balance > ₹7L
Income < ₹20K



Treatment: remove errors • median for missing • log transform

5 COVARIANCE ANALYSIS OF CUSTOMER VARIABLES

•Covariance shows whether two variables move together and indicates the direction of joint variation.

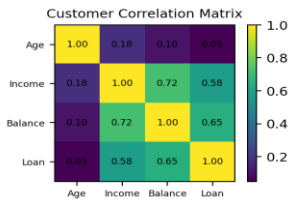


RESULT
Income ↔ Balance: +
Balance ↔ Loan: +

6 CORRELATION ANALYSIS AND CUSTOMER RELATIONSHIPS

•Correlation measures the strength and direction of relationships between customer variables.

1.0 Strong Positive
0 No Correlation
-1.0 Strong Negative

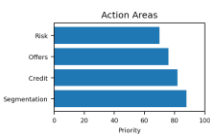


KEY RELATIONSHIPS
Income-Balance ≈ +0.72
Balance-Loan ≈ +0.65

7 BANK CUSTOMER ANALYSIS INSIGHTS AND RECOMMENDATIONS

Correlation and covariance support customer segmentation, credit decisions and targeted banking strategies.

MAJOR INSIGHTS
• Income and balance are positively related
• Outliers need separate review
• Customer segments show different financial patterns



BENEFITS
• Better segmentation
• Smarter credit decisions
• Personalized offers
• Risk identification